

Production Grade Prisma Schema (V1)

Struktur berikut yang saya rekomendasikan sebagai fondasi implementasi:

```
enum Role {  
  CITIZEN  
  OFFICER  
  SUPER_ADMIN  
}
```

```
enum ReportStatus {  
  DRAFT  
  SUBMITTED  
  ASSIGNED  
  IN_PROGRESS  
  RESOLVED  
  CLOSED  
  REJECTED  
}
```

```
enum Priority {  
  LOW  
  MEDIUM  
  HIGH  
  CRITICAL  
}
```

```
enum ReportCategory {  
  INFRASTRUCTURE  
  HEALTH
```

EDUCATION

SANITATION

TRANSPORTATION

OTHER

}

model User {

id String @id @default(uuid())

email String @unique

passwordHash String

fullName String

phone String?

role Role

agencyId String?

agency Agency? @relation(fields: [agencyId], references: [id])

reports Report[] @relation("CitizenReports")

assignedReports Report[] @relation("AssignedReports")

comments ReportComment[]

notifications Notification[]

auditLogs AuditLog[]

isActive Boolean @default(true)

emailVerified Boolean @default(false)

createdAt DateTime @default(now())

```
updatedAt    DateTime @updatedAt
deletedAt    DateTime?
```

```
@@index([role])
```

```
@@index([agencyId])
```

```
}
```

```
model Agency {
```

```
  id      String @id @default(uuid())
```

```
  name    String
```

```
  description String?
```

```
  email   String?
```

```
  phone   String?
```

```
  users   User[]
```

```
  reports Report[]
```

```
  isActive Boolean @default(true)
```

```
  createdAt DateTime @default(now())
```

```
  updatedAt DateTime @updatedAt
```

```
}
```

```
model Location {
```

```
  id      String @id @default(uuid())
```

latitude Decimal

longitude Decimal

address String

district String

city String

province String

postalCode String?

reports Report[]

createdAt DateTime @default(now())

}

model Report {

id String @id @default(uuid())

reportNumber String @unique

title String

description String

category ReportCategory

priority Priority @default(MEDIUM)

status ReportStatus @default(SUBMITTED)

citizenId String

```
citizen      User @relation(  
  "CitizenReports",  
  fields: [citizenId],  
  references: [id]  
)
```

```
agencyId     String  
agency       Agency @relation(  
  fields: [agencyId],  
  references: [id]  
)
```

```
locationId   String  
location     Location @relation(  
  fields: [locationId],  
  references: [id]  
)
```

```
assignedOfficerId String?  
assignedOfficer  User? @relation(  
  "AssignedReports",  
  fields: [assignedOfficerId],  
  references: [id]  
)
```

```
attachments  ReportAttachment[]  
comments     ReportComment[]  
histories    ReportHistory[]
```

resolvedAt DateTime?

closedAt DateTime?

createdAt DateTime @default(now())

updatedAt DateTime @updatedAt

deletedAt DateTime?

@@index([status])

@@index([agencyId])

@@index([citizenId])

@@index([category])

}

model ReportAttachment {

 id String @id @default(uuid())

 reportId String

 report Report @relation(

 fields: [reportId],

 references: [id]

)

 fileName String

 fileUrl String

 fileSize BigInt

 mimeType String

uploadedBy String

createdAt DateTime @default(now())

}

model ReportComment {

id String @id @default(uuid())

reportId String

report Report @relation(

fields: [reportId],

references: [id]

)

userId String

user User @relation(

fields: [userId],

references: [id]

)

message String

createdAt DateTime @default(now())

updatedAt DateTime @updatedAt

}

model ReportHistory {

id String @id @default(uuid())

```
reportId String
report Report @relation(
  fields: [reportId],
  references: [id]
)
```

```
actorId String
actor User @relation(
  fields: [actorId],
  references: [id]
)
```

```
oldStatus ReportStatus?
newStatus ReportStatus
```

```
note String?
```

```
createdAt DateTime @default(now())
}
```

```
model Notification {
  id String @id @default(uuid())
```

```
userId String
user User @relation(
  fields: [userId],
  references: [id]
```


)

title String

message String

isRead Boolean @default(false)

createdAt DateTime @default(now())

}

model AuditLog {

id String @id @default(uuid())

actorId String

actor User @relation(

fields: [actorId],

references: [id]

)

action String

resourceType String

resourceId String

oldValues Json?

newValues Json?

ipAddress String?

```
userAgent String?
```

```
createdAt DateTime @default(now())
```

```
}
```

ENTERPRISE-GRADE NEXT.JS 16 FOLDER STRUCTURE

Ini adalah struktur yang saya rekomendasikan untuk production.

```
src/
```

```
|
```

```
| └─ app/
```

```
| |
```

```
| | └─ (public)/
```

```
| | | └─ page.tsx
```

```
| | | └─ reports/
```

```
| | | └─ about/
```

```
| |
```

```
| | └─ (auth)/
```

```
| | | └─ login/
```

```
| | | └─ register/
```

```
| | | └─ forgot-password/
```

```
| |
```

```
| | └─ (citizen)/
```

```
| | | └─ dashboard/
```

```
| | | └─ reports/
```

```
| | | └─ notifications/
```

```
| | | └─ profile/
```

```
| |
```

- | |— (officer)/
 - | | |— dashboard/
 - | | |— reports/
 - | | |— analytics/
 - | | |— profile/
 - | |
- | |— (admin)/
 - | | |— dashboard/
 - | | |— agencies/
 - | | |— users/
 - | | |— audit-logs/
 - | | |— settings/
 - | |
 - | |— api/
 - | | |— v1/
- | |— modules/
 - | | |— auth/
 - | | | |— components/
 - | | | |— services/
 - | | | |— repositories/
 - | | | |— validators/
 - | | | |— hooks/
 - | | | |— types/
 - | | | |— api/

```
|  ├── users/
|  |
|  ├── agencies/
|  |
|  ├── reports/
|  |
|  ├── attachments/
|  |
|  ├── comments/
|  |
|  ├── notifications/
|  |
|  ├── analytics/
|  |
|  ├── audit-logs/
|  |
|  └── dashboard/
|
|  ├── components/
|  |
|  |  ├── ui/
|  |  |
|  |  ├── forms/
|  |  |
|  |  ├── tables/
|  |  |
|  |  └── charts/
```

```
| |
| |─ maps/
| |
| |─ layouts/
|
|─ server/
|
| |─ auth/
| |
| |─ db/
| |
| |─ cache/
| |
| |─ queue/
| |
| |─ storage/
| |
| |─ telemetry/
| |
| |─ permissions/
|
|─ lib/
|
| |─ constants/
| |
| |─ env/
| |
```

```
|   ├── logger/
|   |
|   ├── utils/
|   |
|   └── validators/
|
|   ├── providers/
|   |
|   |   ├── query-provider.tsx
|   |   ├── theme-provider.tsx
|   |   └── auth-provider.tsx
|   |
|   |   ├── prisma/
|   |   |
|   |   |   ├── schema.prisma
|   |   |   ├── migrations/
|   |   |   └── seed.ts
|   |   |
|   |   ├── tests/
|   |   |
|   |   |   ├── unit/
|   |   |   ├── integration/
|   |   |   └── e2e/
|   |   |
|   |   ├── types/
|   |   |
|   |   └── middleware.ts
```

```
|  
└─ instrumentation.ts
```

Struktur Enterprise Internal Module

Contoh untuk module reports.

```
modules/  
└─ reports/  
    |  
    └─ components/  
        |  
        └─ services/  
            | └─ create-report.service.ts  
            | └─ assign-report.service.ts  
            | └─ update-status.service.ts  
            | └─ resolve-report.service.ts  
            |  
            └─ repositories/  
                | └─ report.repository.ts  
                |  
                └─ validators/  
                    | └─ create-report.schema.ts  
                    | └─ update-report.schema.ts  
                    |  
                    └─ types/  
                        |  
                        └─ hooks/  
                            |
```

|— api/

|

|— constants/